Publications - Felipe Alarcón

Complete publication list on NASA ADS

54	9 + 5	27	2,590
Refereed publications	First / second author	h-index	Citations (NASA ADS)

Plus 7 submitted articles under review. † details of personal contribution.

First-author papers

1. "EK Cha b: multi-tracer detection of a protoplanet candidate in a gas-rich environment." **Alarcón**, Chowbay, Facchini et al, A&A, 715, L10 (2026).
2. "exoALMA. XXIV. Formaldehyde Emission in Protoplanetary Disks of exoALMA Compared with Their Properties and Dynamical State." **Alarcón**, Facchini, Trapman et al, ApJL 1000 L32 (2026).
3. "Silicon Sulfide Isotopologues Reveal Silicate Sublimation in the Outer Disk of PDS 66" Yoshida, **Alarcón** et al, ApJL, 999, L22 (2026). *(shared first authorship with Yoshida)*
4. "Thermal Structure and Millimeter Emission of Protoplanetary Disk with embedded protoplanets from radiative transfer modeling" **Alarcón** & Bergin, ApJ, 967, 144 (2024).
5. "Extinction Values toward Embedded Planets in Protoplanetary Disks Estimated from Hydrodynamic Simulations" **Alarcón**, Bergin, and Cugno, ApJ, 966, 225 (2024)
6. "Viscous heating as the dominant heat source inside the water snowline of V883 Ori" **Alarcón**, Casassus, Lyra et al, MNRAS, 527, 9655 (2024).
7. "A Localized Kinematic Structure Detected in Atomic Carbon Emission Spatially Coincident with a Proposed Protoplanet in the HD 163296 Disk" **Alarcón**, Bergin, and Teague ApJL, 941, L24 (2022)
8. "Molecules with ALMA at Planet-forming Scales (MAPS). VIII. CO Gap in AS 209-Gas Depletion or Chemical Processing?" **Alarcón**, Bosman, Bergin et al. ApJS, 257, 8 (2021)
9. "Chemical Evolution in a Protoplanetary Disk within Planet Carved Gaps and Dust Rings" **Alarcón**, Teague, Zhang et al. ApJ, 905, 68 (2020)

Refereed papers · major or significant contributions

* denotes second authorship.

10. ***** "Dust Recycling in Protoplanetary Disks: A Novel Method of Volatile Enrichment in Embedded Giant Planets" Van Clepper, **Alarcón**, Ciesla et al. ApJL, 994, L44 (2025).
11. ***** "Tracking the Chemical Evolution of Hydrocarbons Through Carbon Grain Supply in Protoplanetary Disks" Raul, **Alarcón**, and Bergin ApJ, 982, 155 (2025)
12. ***** "CI Traces the Disk Atmosphere in the IM Lup Protoplanetary Disk" Law, **Alarcón**, Cleeves et al. ApJL, 959, L27 (2023)
13. ***** "Molecules with ALMA at Planet-forming Scales (MAPS). VII. Substellar O/H and C/H and Superstellar C/O in Planet-feeding Gas" Bosman, **Alarcón**, Bergin et al. ApJS, 257, 7 (2021)
14. ***** "Destruction of Refractory Carbon Grains Drives the Final Stage of Protoplanetary Disk Chemistry" Bosman, **Alarcón**, Zhang et al. ApJ, 910, 3 (2021)
15. "Direct Measurement of Extinction in a Planet-Hosting Gap " Cugno, Facchini, **Alarcón** et al. AJ, 170, 317 (2025)
16. "PDRs4All. VIII. Mid-infrared emission line inventory of the Orion Bar" Van De Putte, Meshaka, Trahin et al. A&A, 687, A86 (2024)
17. "A far-ultraviolet–driven photoevaporation flow observed in a protoplanetary disk" Berné, Habart, Peeters et al. Sci, 383, 988 (2024)
18. "Formation of the methyl cation by photochemistry in a protoplanetary disk" Berné, Martin-Drumel, Schroetter et al. Nature, 621, 56 (2023)

19 "PDRs4All. IV. An embarrassment of riches: Aromatic infrared bands in the Orion Bar" Chown, Sidhu, Peeters et al. A&A, 685, A75 (2024)

20 "Molecules with ALMA at Planet-forming Scales. XX. The Massive Disk around GM Aurigae" Schwarz, Calahan, Zhang et al. ApJS, 257, 20 (2021)

Refereed papers · author contributions and/or data reduction

21 "ALMA detection of continuum emission from GQ Lup B" Pirola, Facchini, Cugno et al., accepted for pblication in A&A.

22 "JWST-DECO: The Impact of Accretion on Mid-Infrared Observable Water in Planet-forming Disks." Calahan, Dziire, Öberg et al. ApJ, 1004,213 (2026)

23 "PDRs4All: XX. The 6 to 9μm region as a probe of PAH charge and size in the Orion Bar." Khan, Daza Rodriguez, Peeters et al. A&A, 709, A39 (2026)

24 "PDRs4All: XIX. The evolution of the PAH ionisation and PAH size distribution across the Orion Bar." Maragkoudakis, Boersma, Peeters et al. A&A, 709, A38 (2026)

25 "Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement" Armitage, Williams, Zhang et al. ApJ, 998, 308 (2026)

26 "JWST-MIRI Observations of the Irradiated Chemistry in the Inner Disk Cavity of GM Aur." Romero-Mirza, Öberg, Banzatti et al. ApJ, 991, 128 (2025)

27 "PDRs4All XV: CH radical and H_3^+ molecular ion in the irradiated protoplanetary disk d203-506" Schroetter, Berné, Goicoechea et al. A&A, 699, L13 (2025)

28 "PDRs4All: XIV. Probing CH out-of-plane bending modes of PAH molecules in the Orion Bar with JWST" Khan, Abbott, Peeters et al. A&A, 699, A133 (2025)

29 "A solar C/O ratio in planet-forming gas at 1 au in a highly irradiated disk" Schroetter, Berné, Bron et al. Nat Astron (2025). doi:10.1038/s41550-025-02596-6

30 "PDRs4All: XIII. Empirical prescriptions for the interpretation of JWST imaging observations of star-forming regions" Chown, Okada, Peeters et al. A&A, 698, A86 (2025)

31 "PDRs4All: XII. Far-ultraviolet-driven formation of simple hydrocarbon radicals and their relation with polycyclic aromatic hydrocarbons" Goicoechea, Pety, Cuadrado et al. A&A, 696, A100 (2025)

32 "PDRs4All: X. ALMA and JWST detection of neutral carbon in the externally irradiated disk d203-506: Undepleted gas-phase carbon" Goicoechea, Le Bourlot, Black et al. A&A, 689, L4 (2024)

33 "PDRs4All. IX. Sulfur elemental abundance in the Orion Bar" Fuente, Roueff, Le Petit et al. A&A, 687, A87 (2024)

34 "PDRs4All. VII. The 3.3 μm aromatic infrared band as a tracer of physical properties of the interstellar medium in galaxies" Schroetter, Berné, Joblin et al. A&A, 685, A78 (2024)

35 "PDRs4All. VI. Probing the photochemical evolution of PAHs in the Orion Bar using machine learning techniques" Pasquini, Peeters, Schefter et al. A&A, 685, A77 (2024)

36 "OH as a probe of the warm-water cycle in planet-forming disks" Zannese, Tabone, Habart et al. NatAs, 8, 577 (2024)

37 "Astronomical CH3+ rovibrational assignments. A combined theoretical and experimental study validating observational findings in the d203-506 UV-irradiated protoplanetary disk" Changala, Chen, Li et al. A&A, 680, A19.

38 "PDRs4All. V. Modelling the dust evolution across the illuminated edge of the Orion Bar" Elyajouri, Ysard, Abergel et al. A&A, 685, A76 (2024)

39 "PDRs4All: III. JWST's NIR spectroscopic view of the Orion Bar" Peeters, Habart, Berné et al. A&A, 685, A74 (2024)

40 "PDRs4All. II. JWST's NIR and MIR imaging view of the Orion Nebula" Habart, Peeters, Berné et al. A&A, 685, A73 (2024)

41 "Molecules with ALMA at Planet-forming Scales (MAPS): Complex Kinematics in the AS 209 Disk Induced by a Forming Planet and Disk Winds" Galloway-Sprietsma, Bae, Teague et al. ApJ, 950, 147 (2023)

42 "Molecules with ALMA at Planet-forming Scales (MAPS): A Circumplanetary Disk Candidate in Molecular-line Emission in the AS 209 Disk" Bae, Teague, Andrews et al. ApJL, 934, L20 (2022)

43 "PDRs4All: A JWST Early Release Science Program on Radiative Feedback from Massive Stars" Berné, Habart, Peeters et al. PASP, 134, 054301 (2022)

44 "Molecules with ALMA at Planet-forming Scales (MAPS). I. Program Overview and Highlights" Öberg, Guzmán, Walsh et al. ApJS, 257, 1 (2021)

45 "Molecules with ALMA at Planet-forming Scales (MAPS). XVII. Determining the 2D Thermal Structure of the HD 163296 Disk" Calahan, Bergin, Zhang et al. ApJS, 257, 17 (2021)

46 "Molecules with ALMA at Planet-forming Scales (MAPS). XV. Tracing Protoplanetary Disk Structure within 20 au" Bosman, Bergin, Loomis et al. ApJS, 257, 15 (2021)

47 "Molecules with ALMA at Planet-forming Scales (MAPS). IV. Emission Surfaces and Vertical Distribution of Molecules" Law, Teague, Loomis et al. ApJS, 257, 4 (2021)

48 "Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions" Zhang, Booth, Law et al. ApJS, 257, 5 (2021)

49 "Molecules with ALMA at Planet-forming Scales (MAPS). VI. Distribution of the Small Organics HCN, C_2H, and H_2CO" Guzmán, Bergner, Law et al. ApJS, 257, 6 (2021)

50 "Molecules with ALMA at Planet-forming Scales (MAPS). III. Characteristics of Radial Chemical Substructures" Law, Loomis, Teague et al. ApJS, 257, 3 (2021)

51 "Long Baseline Observations of the HD 100546 Protoplanetary Disk with ALMA" Pérez, Casassus, Hales et al. ApJL, 889, L24 (2020)

52 "Dust traps in the protoplanetary disc MWC 758: two vortices produced by two giant planets?" Baruteau, Barraza, Pérez et al. MNRAS, 486, 304 (2019)

53 "Cm-wavelength observations of MWC 758: resolved dust trapping in a vortex" Casassus, Marino, Lyra et al. MNRAS, 483, 3278 (2019)

54 "An inner warp in the DoAr 44 T Tauri transition disc" Casassus, Avenhaus, Pérez et al. MNRAS, 477, 5104 (2018)

Submitted articles

55 "DEC/O: Benchmarking Protoplanetary Disk Chemistry Codes for C/H and C/O Retrievals" Anderson, Walsh, Aikawa et al., submitted to A&A.

56 "DEC/O I: A Chemical Census of 80 Planet-Forming Disks" Cleeves, Anderson, Guzmán et al., submitted to A&A.

57 "DEC/O IV. CO Visibility Modeling of Protoplanetary Disks: Gas Disk Sizes, Stellar Ages, and Radial Drift Signatures" Zallio, Rosotti, Vioque et al., submitted to A&A.

58 "The Disk-Exoplanet C/Onnection (DEC/O). VI. An Overview of Line Emission Radial Structure" Law, Cleeves, Green et al., submitted to A&A.

59 "DEC/O X: JWST and ALMA Reveal a Connection Between Inner- and Outer-Disk Carbon Chemistry" Colmenares, Bergin, Cleeves et al., submitted to A&A .

60 "DEC/O XI: Characterizing Formaldehyde (H2CO) emission and excitation properties in protoplanetary disks" Hernández-Vera, Guzmán, Miotello et al., submitted to A&A.

61 "Slowly Spiraling CQ Tau: Constraints on Disk Pattern Drivers and Accreting Companions" Tobin, Cugno, Meyer et al., submitted to ApJ.